

NUO (IVY) LIU, PH.D.

Postdoctoral Fellow | Goodarzi Lab, Arc Institute & University of California, San Francisco
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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA
Ph.D., Computational and Systems Biology | GPA: 4.9/5.0
Advisor: Prof. Alex K. Shalek (Chemistry, IMES, HST)

Harvey Mudd College Claremont, CA
B.S., Mathematical and Computational Biology | GPA: 3.95/4.0
Graduated with High Distinction | Honors in Biology | Honors in Computer Science
Humanities concentration in Economics | Dean's List all applicable semesters

Honors & Awards

David H. Koch Graduate Fellowship, Koch Institute, MIT	2023–24
Computer Science Class of '94 Award, Harvey Mudd College	2020
CRA Outstanding Undergraduate Research Award, Finalist	2020
W.A. Brandenburger Biology Prize, Harvey Mudd College	2019
First-Tier Traveling Fellowship, 17th Asia Pacific Bioinformatics Conference	2019
CRC Handbook Prize, Harvey Mudd College	2017

Research Experience

Arc Institute & UCSF — Postdoctoral Fellow, Goodarzi Lab Aug 2025 – Present
Advisor: Prof. Hani Goodarzi (Arc/UCSF)

- Developing cancer-specialized RNA foundation models for oncRNA target discovery across 32 cancer types
- Building hybrid computational-experimental pipelines for RNA 3D structure prediction
- Engineering AI-driven synthetic RNA-binding protein (RBP) switches for precision targeting of cancer-specific oncRNAs
- Developing multimodal biological reasoning LLMs for protein function prediction (BioReason-Pro)
- Building a Gene × Environment (G×E) map using multiplexed single-cell screening across 40+ cancer cell lines in 96 metabolic perturbation conditions; developing the STATE virtual cell model to predict cancer metabolic vulnerabilities in silico (Perturb-Env; in collaboration with Jain Lab & Tahoe Therapeutics)
- Characterizing APOE genetic variant effects on Alzheimer's disease using human ex vivo models (in collaboration with Konermann Lab)

Massachusetts Institute of Technology — Graduate Researcher, Shalek Lab Jun 2021 – Aug 2025

Broad Institute & Ragon Institute, Cambridge, MA

- Pioneered compressed phenotypic screening: a broadly applicable experimental and computational framework for scaling phenotypic screens on ex vivo multicellular models; discovered pleiotropic effects of JAK-STAT inhibitors on human immune cells
- Led single-cell and spatial transcriptomic profiling of human TB-diseased lung tissue; identified myofibroblast-like populations as central drivers of TB pathogenesis and lung remodeling; work redefined TB fibrosis biology
- Used single-cell transcriptomics to characterize functional mechanisms of long-term growth hormone administration effects on MASLD/NAFLD patients

Westlake University — Summer Research Intern, Guo Lab

Jul – Aug 2020

Hangzhou, China | Advisor: Dr. Tiannan Guo

- Built and trained an LSTM-based deep learning model on pan-human tissue proteomics to predict ion mobility values (Pearson's $r > 0.98$)
- Predicted ion mobility values for mouse liver proteomics data from non-timsTOF instruments; compared library search results using ion mobility-supplemented libraries

Harvey Mudd College — Senior Thesis Researcher, Bush Lab

Sep 2019 – May 2020

Claremont, CA | Advisor: Prof. Eliot Bush

- Year-long thesis: Reconstructing Gene Family Evolution in Microbes Using the DTLOr Algorithm
- Extended Duplication-Transfer-Loss model to include new syntenic events for more comprehensive reconstruction of prokaryotic evolutionary history

Baylor College of Medicine — Summer Research Intern, Sumazin Lab

Summer 2019

Houston, TX | Advisor: Dr. Pavel Sumazin

- Designed a method to estimate cell-type specific expression from bulk tumor profiles using a hierarchical Linear Programming approach to reconstruct transcriptional evolution of tumors

Baylor College of Medicine — Summer Research Intern, Coarfa Lab

Summer 2018

Houston, TX | Advisor: Dr. Cristian Coarfa

- Performed RNA-seq, microarray, and gene enrichment analysis on rat liver epigenome under environmental influences
- Built and trained a CNN to predict epigenetic remarking events; extracted de novo DNA motifs related to epigenetic reprogramming

Harvey Mudd College — Research Intern, Libeskind-Hadas & Wu Labs

Summer 2017

Claremont, CA

- Extended dynamic programming algorithm for phylogenetic tree reconciliation under the Duplication-Loss-Coalescence model to optimize Pareto-optimal solutions

Harvey Mudd College — Research Intern, McFadden Lab

Spring 2017

Claremont, CA | Advisor: Prof. Catherine McFadden

- Investigated feasibility of 28S rDNA genetic barcoding to differentiate species within coral genus *Sinularia*; demonstrated promise as an alternative to whole-genome sequencing

Publications & Preprints

* Co-first authorship

2026

Fallahpour, A., Seyed-Ahmadi, A., Idehpour, P., Ibrahim, O., Gupta, P., Naimer, J., Zhu, K., Shah, A., Ma, S., Adduri, A., Güloğlu, T., **Liu, N.**, Cui, H., Jain, A., de Castro, M., Stiles, J.S., Nemcko, F., Nevue, A.A., Moon, H.C., Sosnick, L., Markham, O., Duan, H., Lee, M.Y.Y., Salvador, A.F.M., Maddison, C.J., Thaïss, C.A., Ricci-Tam, C., Plosky, B.S., Burke, D.P., Hsu, P.D., Goodarzi, H., Wang, B. (2026). BioReason-Pro: Advancing Protein Function Prediction with Multimodal Biological Reasoning. *bioRxiv* 10.64898/2026.03.19.712954.

Liu, N.*, Mbano, I.M.*, Wadsworth, M.H., Chambers, M.J., Mpotje, T., Asowata, O.E., Nyquist, S.K., Nargan, K., Ramsuran, D., Karim, F., Hughes, T., Bromley, J.D., Krause, R., Naidoo, T., Tezera, L., Reichmann, M.T., Ganchua, S.K., Kløverpris, H.N., Dullabh, K.J., ... Leslie, A. (2026). Single-cell and spatial profiling highlights TB-induced myofibroblasts as drivers of lung pathology. *J. Exp. Med.* 223(3), e20251067. <https://doi.org/10.1084/jem.20251067>

2025

Liu, N.*, Kattan, W.E.*, Mead, B.E.*, Kummerlowe, C.*, Cheng, T., Cheah, J.H., Soule, C.K., Peters, J., Lowder, K.E., Blainey, P.C., Hahn, W.C., Cleary, B., Bryson, B., Winter, P.S., Raghavan, S., & Shalek, A.K. (2025). Compressed phenotypic screens for complex multicellular models and high-content assays. *Nature Biotechnology* 43, 1324–1336. <https://doi.org/10.1038/s41587-024-02403-z>

2023

Liu, N., Gonzalez, T.A., Fischer, J., Hong, C., Johnson, M., Mawhorter, R., Mugnatto, F., Soh, R., Somji, S., Wirth, J.S., Libeskind-Hadas, R., & Bush, E.C. (2023). xenoGI 3: Using the DTOR model to reconstruct the evolution of gene families in clades of microbes. *BMC Bioinformatics* 24, 295. <https://doi.org/10.1186/s12859-023-05410-0>

2021

Raghavan, S., Winter, P.S., Navia, A.W., Williams, H.L., DenAdel, A., Lowder, K.E., Galvez-Reyes, J., Kalekar, R.L., Mulugeta, N., Kapner, K.S., Raghavan, M.S., Borah, A.A., **Liu, N.**, Väyrynen, S.A., Costa, A.D., Ng, R.W.S., Wang, J., Hill, E.K., Ragon, D.Y., ... Shalek, A.K. (2021). Microenvironment drives cell state, plasticity, and drug response in pancreatic cancer. *Cell* 184(25), 6119–6137. <https://doi.org/10.1016/j.cell.2021.11.017>

Du, H., Ong, Y.S., Knittel, M., Mawhorter, R., **Liu, N.**, Gross, G., Tojo, R., Libeskind-Hadas, R., & Wu, Y.-C. (2021). Multiple Optimal Reconciliations Under the Duplication-Loss-Coalescence Model. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 18, 2144–2156. <https://doi.org/10.1109/TCBB.2019.2922337>

Liu, J., Mawhorter, R., **Liu, N.**, Santichaivekin, S., Bush, E., & Libeskind-Hadas, R. (2021). Maximum parsimony reconciliation in the DTOR model. *BMC Bioinformatics* 22, 394. <https://doi.org/10.1186/s12859-021-04290-6>

2020

Carothers, M., Gardi, J., Gross, G., Kuze, T., **Liu, N.**, Plunkett, F., Qian, J., & Wu, Y.-C. (2020). An Integer Linear Programming Solution for the Most Parsimonious Reconciliation Problem under the Duplication-Loss-Coalescence Model. *ACM BCB 2020*. <https://doi.org/10.1145/3388440.3412474>

2019

Mawhorter, R., **Liu, N.**, Libeskind-Hadas, R., & Wu, Y.-C. (2019). Inferring Pareto-optimal reconciliations across multiple event costs under the duplication-loss-coalescence model. *BMC Bioinformatics* 20, 639. <https://doi.org/10.1186/s12859-019-3206-6>

Selected Presentations

Lightning Talk (Nominated): “Perturb-Env: Towards a GxE map for virtual cells”	Feb 2026
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Arc Institute Symposium, San Francisco, CA

Oral: Compressed phenotypic screening empowers scalable biological discovery	May 2024
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Single-Cell Genomics Conference GRC, Les Diablerets, Switzerland

Talk: Compressed Screening — High-throughput measurement and perturbation of tissues	Feb 2024
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Microsoft Research New England, Cambridge, MA

Talk: Putative drivers of cellular plasticity in glioblastoma	Jul 2022
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NCI Human Tumor Atlas Network (HTAN) Internal Meeting, Cambridge, MA

Talk: Tumor phylogeny inference from gene expression profiles of multiple tumor mixtures Jul 2019

Baylor College of Medicine SMART Program Final Symposium, Houston, TX

Oral: Multiple Optimal Reconciliations under the DLC Model Jan 2019

17th Asia Pacific Bioinformatics Conference, Wuhan, China

Teaching & Service

Teaching Assistant — Modern Biostatistics (7.093) & Modern Computational Biology (7.094), MIT Spring 2022

Designed and led weekly recitations and office hours; co-developed course materials; graded problem sets and exams

Teacher — MIT Spark 2022 (Immunology for 7th & 8th graders) Mar 2022

Workshop Organizer — Single-Cell Genomics Workshop in Asia, Human Cell Atlas 2024

Executive Board (Secretary & Fund Manager) — CSB Application Assistance Program, MIT 2021 – 2025

Helped establish the program; secured Grad Student Experience Grant from MIT Office of Graduate Education

Co-President & Director of Planning — MIT-CHIEF (China Innovation & Entrepreneurship Forum) 2020 – 2025

Mentor — Graduate students, undergraduate URAs, visiting students, Shalek Lab 2021 – 2025

Mentored international collaborators (Human Cell Atlas, Child Health Research Foundation Bangladesh) on single-cell analysis

Head Mentor — APISPAM (Asian Pacific Islander Sponsor Program), Harvey Mudd College 2017 – 2020

Recruited and trained 20+ mentors; organized retreat and year-long cultural events

Tutor & Grader — Harvey Mudd College (Discrete Math, CompBio, CS, ML, Databases) 2017 – 2020

Academic Excellence Program Facilitator — Harvey Mudd College Biology 2019 – 2020